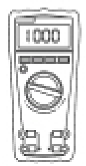
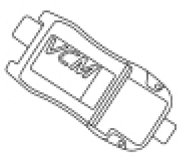
Stoplamps

Ford Ecat Electronic Parts Catalog

Special Tool(s)

 ST3093-A	Fluke 77-1V Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

The Body Control Module (BCM) monitors the input from the stoplamp switch. When the brake pedal is applied, voltage is routed to the BCM and the PCM. The BCM then supplies voltage to the stoplamps.

Aside from the stoplamp operation, when the brake pedal is applied, voltage passes through the BCM fuse 31 (5A) to the Trailer Brake Control (TBC) module and a separate circuit for customer use.

Field-Effect Transistor (FET) Protection

A Field-Effect Transistor (FET) is a type of transistor that, when used with module software, monitors and controls current flow on module outputs. The FET protection strategy prevents module damage in the event of excessive current flow.

The BCM utilizes a FET protective circuit strategy for many of its outputs (such as a headlamp output circuit). Output loads (current level) are monitored for excessive current (typically short circuits) and are shut down when a fault event is detected. A short circuit DTC is stored at the fault event and a cumulative counter is started.

When the demand for the output is no longer present, the module resets the FET protection, allowing the circuit to function. If the circuit is still shorted the next time the driver requests a circuit to activate that has been shut down by a previous short (FET protection), the FET protection shuts off the circuit again and the cumulative counter advances.

When the excessive circuit load occurs often enough, the module shuts down the output until a repair procedure is carried out. Each FET protected circuit has 3 predefined levels of short circuit tolerance based on the harmful effect of each circuit fault on the FET and the ability of the FET to withstand it. A module lifetime level of fault events is established based upon the durability of the FET . If the total tolerance level is determined to be 600 fault events, the 3 predefined levels would be 200, 400 and 600 fault events.

When a tolerance level is reached, the short circuit DTC that was stored on the first failure cannot be cleared by the clear the continuous DTCs command. The module does not allow this code to be cleared or the circuit restored to normal operation until a successful self-test proves that the fault has been repaired. After the self-test has successfully completed (no on-demand DTCs present), DTC U1000:00 and the associated DTC (the DTC related to the shorted circuit) automatically clears and the circuit function returns. The module never resets the fault event counter to zero and continues to advance the fault event counter as short circuit fault events occur.

If the number of short circuit fault events reach the third level, then DTCs U1000:00 and U3000:49 set along with the associated short circuit DTC. DTC U3000:49 cannot be cleared and the module must be replaced after the repair.

The BCM FET protected output circuits for the stoplamps system are the high mounted stoplamp, and the LH and RH rear stoplamp output circuits.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Stoplamp switch	• Battery Junction Box (BJB) fuse 46 (10A) (stoplamp switch) • Body Control Module (BCM) fuse(s): ▪ 13 (15A) (RH rear stoplamp) ▪ 14 (15A) (LH rear stoplamp) ▪ 15 (15A) (high mounted stoplamp) • Bulb • Bulb socket • Wiring, terminals or connectors • Rear lamp assembly • High mounted stoplamp

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. **NOTE:** *Make sure to use the latest scan tool software release.*

   If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .
5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

   If the scan tool does not communicate with the VCM :
   - check the VCM connection to the vehicle.
   - check the scan tool connection to the VCM .
   - refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.
6. If the scan tool does not communicate with the vehicle:
   - verify the ignition is on.
     - The air bag warning indicator prove-out confirms ignition on (other indicators may not prove ignition on). If the ignition does not turn on, refer to Section 211-05 to diagnose no power in run.
   - verify the scan tool operation with a known good vehicle.
   - refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test.
    - If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
    - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the BCM .
9. If the DTCs retrieved are related to the concern, refer to Diagnostic Trouble Code (DTC) Chart in this section. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.
10. If no DTCs related to the concern are retrieved, GO to Symptom Chart.

Symptom Chart

Symptom Chart

Condition	Possible Causes	Action
• All the stoplamps are inoperative	• Fuse • Wiring, terminals or connectors • Stoplamp switch • Body Control Module (BCM)	• VERIFY the bulbs are OK. If OK, GO to Pinpoint Test I.
• One or more stoplamps are inoperative	• Fuse • Wiring, terminals or connectors • Bulb socket • High mounted stoplamp • BCM	• VERIFY the bulbs are OK. If OK, GO to Pinpoint Test J.
• The stoplamp(s) are on continuously	• Wiring, terminals or connectors • Stoplamp switch • Trailer Brake Control (TBC) module • PCM • BCM	• GO to Pinpoint Test K.

Pinpoint Tests

Pinpoint Test I: All The Stoplamps Are Inoperative

Refer to Wiring Diagrams Cell 90 , Turn Signal/Stop/Hazard Lamps for schematic and connector information.

Normal Operation

The stoplamp switch is supplied voltage from the Battery Junction Box (BJB) fuse 46 (10A). When the brake pedal is applied, the stoplamp switch routes voltage to the Body Control Module (BCM) , indicating the brake pedal is applied.

The BCM provides voltage to all the stoplamps (on independent circuits) when it detects the brake pedal is applied.

This pinpoint test is intended to diagnose the following:

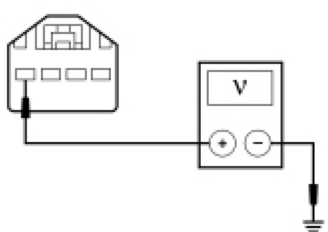
- Fuse
- Wiring, terminals or connectors
- Stoplamp switch
- BCM

PINPOINT TEST I : ALL THE STOPLAMPS ARE INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

I1 CHECK FOR VOLTAGE TO THE STOPLAMP SWITCH

- Ignition OFF.
- Disconnect: Stoplamp Switch C278.
- Measure the voltage between the stoplamp switch C278 Pin 4, circuit SBB46 (BN/RD), harness side and ground.



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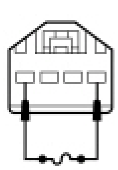
Is the voltage greater than 10 volts?

Yes	GO to I2.
No	VERIFY the BJB fuse 46 (10A) is OK. If OK, REPAIR circuit SBB46 (BN/RD) for an open. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short.

I2 BYPASS THE STOPLAMP SWITCH

- Connect a fused jumper wire between the stoplamp switch C278 Pin 4, circuit SBB46 (BN/RD), harness side and the stoplamp switch C278 Pin 1, circuit CCB08 (VT/WH), harness side.



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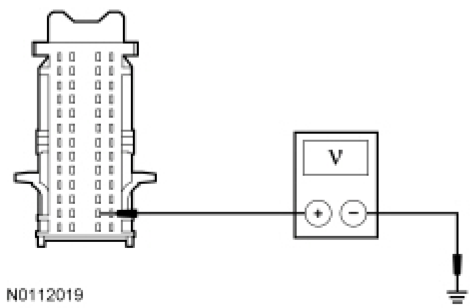
Are the stoplamps illuminated?

Yes	REMOVE the jumper wire. INSTALL a new stoplamp switch. REFER to Stoplamp Switch in this section. TEST the system for normal operation.
No	LEAVE the jumper wire connected. GO to I3.

I3 CHECK THE STOPLAMP SWITCH SIGNAL TO THE BCM

- Disconnect: BCM C2280B .
- Measure the voltage between the BCM C2280B Pin 28, circuit CCB08 (VT/WH), harness side and ground.
- 

Is the voltage greater than 10 volts?

Yes	REMOVE the jumper wire. GO to I4.
No	REMOVE the jumper wire. REPAIR circuit CCB08 (VT/WH) for an open. TEST the system for normal operation.

I4 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test J: One Or More Stoplamps Are Inoperative

Refer to Wiring Diagrams Cell 90 , Turn Signal/Stop/Hazard Lamps for schematic and connector information.

Normal Operation

When the brake pedal is applied, the stoplamp switch routes voltage to the Body Control Module (BCM) , indicating the brake pedal is applied.

The BCM provides voltage to all the stoplamps (on independent circuits) when it detects the brake pedal is applied.

The LH stoplamps share the fused power source with the LH front turn lamp and the LH exterior mirror turn lamp.

The RH stoplamps share the fused power source with the RH front turn lamp and the RH exterior mirror turn lamp.

The high mounted stoplamp shares the fused power source with the reversing lamps.

The rear stoplamps share a common ground circuit.

DTC Description	Fault Trigger Conditions
• B1115:11 — High Mounted Stop Lamp Control: Circuit Short To Ground	This DTC sets when the BCM detects a short to ground from the high mounted stoplamp voltage supply circuit.
• B132B:11 — Left Stop/Turn Lamp: Circuit Short To Ground	This DTC sets when the BCM detects a short to ground from the LH rear stop/turn lamp voltage supply circuit.
• B132B:15 — Left Stop/Turn Lamp: Circuit Short To Battery or Open	This DTC sets when the BCM detects an open from the LH rear stop/turn lamp voltage supply circuit.
• B132C:11 — Right Stop/Turn Lamp: Circuit Short To Ground	This DTC sets when the BCM detects a short to ground from the RH rear stop/turn lamp voltage supply circuit.
• B132C:15 — Right Stop/Turn Lamp: Circuit Short To Battery or Open	This DTC sets when the BCM detects an open from the RH rear stop/turn lamp voltage supply circuit.
• U1000:00 — Solid State Driver Protection Active - Driver Disabled: No Sub Type Information	This DTC sets when the BCM has temporarily shut down the output driver. **The module has temporarily disabled a stoplamp output because an excessive current draw exists (such as a short to ground). The BCM cannot enable the stoplamp output until the cause of the short is corrected, the DTCs have been cleared and a successful self-test is run.**
• U3000:49 — Control Module: Internal Electronic Failure	This DTC sets when the BCM has permanently shut down the output driver. **The module has permanently disabled a stoplamp output because an excessive current draw fault (such as a short to ground) has exceeded the limits that the BCM can withstand. CORRECT the cause of the excessive current draw before installing a new BCM .**

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Bulb socket
- High mounted stoplamp
- BCM

PINPOINT TEST J : ONE OR MORE STOPLAMPS ARE INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Make sure the bulb is good before continuing diagnostics.*

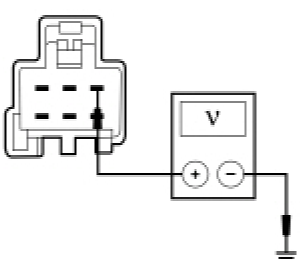
J1 DETERMINE THE INOPERATIVE STOPLAMP

- Ignition OFF.
- Apply the brake pedal and observe the stoplamps.

Are all the stoplamps inoperative?

Yes	GO to Pinpoint Test I.
No	For the high mounted stoplamp, GO to J2. If an individual rear stoplamp is inoperative, GO to J7. If both LH and RH rear stoplamps are inoperative, REPAIR circuit GD117 (BK) for an open. TEST the system for normal operation.

J2 CHECK FOR VOLTAGE TO THE HIGH MOUNTED STOPLAMP

- Disconnect: High Mounted Stoplamp C904.
- While applying the brake pedal, measure the voltage between the high mounted stoplamp C904 Pin 3, circuit CLS52 (BN), harness side and ground.
- 

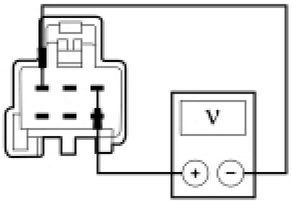
N0117374

Is the voltage greater than 10 volts?

Yes	GO to J3.
No	VERIFY the BCM fuse 15 (15A) is OK. If OK, GO to J4. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. If no DTCs are present, TEST the system for normal operation. If DTC U1000:00 is present, CLEAR the DTCs **and** REPEAT the self-test **(required to enable the stoplamp output driver if DTC U1000:00 is present).** TEST the system for normal operation. If DTC U3000:49 is present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

J3 CHECK THE HIGH MOUNTED STOPLAMP GROUND CIRCUIT FOR AN OPEN

- While applying the brake pedal, measure the voltage between the high mounted stoplamp C904 Pin 3, circuit CLS52 (BN), harness side and the high mounted stoplamp C904 Pin 1, circuit GD139 (BK/YE), harness side.



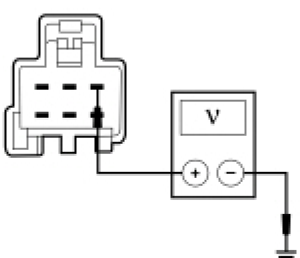
- N0117376

Is the voltage greater than 10 volts?

Yes	INSTALL a new high mounted stoplamp. TEST the system for normal operation.
No	REPAIR circuit GD139 (BK/YE) for an open. TEST the system for normal operation.

J4 CARRY OUT THE BCM ON-DEMAND SELF-TEST AND RECHECK FOR VOLTAGE TO THE HIGH MOUNTED STOPLAMP

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: BCM Self-Test .
- Clear the DTCs and repeat the BCM on-demand self-test.
- While applying the brake pedal, measure the voltage between the high mounted stoplamp C904 Pin 3, circuit CLS52 (BN), harness side and ground.



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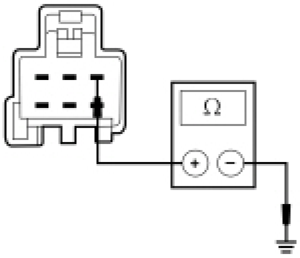
Is the voltage greater than 10 volts?

Yes	INSTALL a new high mounted stoplamp. TEST the system for normal operation.
No	GO to J5.

J5 CHECK THE HIGH MOUNTED STOPLAMP OUTPUT CIRCUIT FOR A SHORT TO GROUND

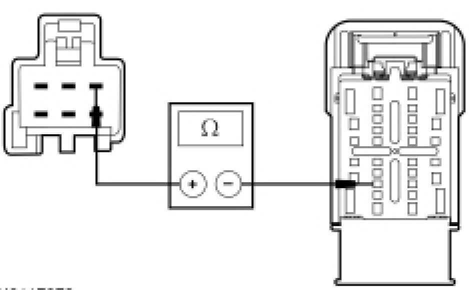
- Ignition OFF.

- Disconnect: BCM C2280D .
- Measure the resistance between the high mounted stoplamp C904 Pin 3, circuit CLS52 (BN), harness side and ground.
- 
  N0117377

Is the resistance greater than 10,000 ohms?

Yes	GO to J6.
No	REPAIR circuit CLS52 (BN) for a short to ground. After the repair: If no DTCs are present, TEST the system for normal operation. If DTC U1000:00 is present, CLEAR the DTCs **and** REPEAT the self-test **(required to enable the stoplamp output driver if DTC U1000:00 is present).** TEST the system for normal operation. If DTC U3000:49 is present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

J6 CHECK THE HIGH MOUNTED STOPLAMP OUTPUT CIRCUIT FOR AN OPEN

- Measure the resistance between the high mounted stoplamp C904 Pin 3, circuit CLS52 (BN), harness side and the BCM C2280D Pin 14, circuit CLS52 (BN), harness side.
- 
  N0117378

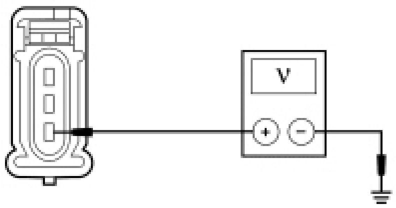
Is the resistance less than 5 ohms?

Yes	GO to J12.
No	REPAIR circuit CLS52 (BN) for an open. TEST the system for normal operation.

J7 CHECK FOR VOLTAGE TO THE REAR STOPLAMP

- Disconnect: Inoperative Rear Lamp .
- While applying the brake pedal, measure the voltage between the inoperative lamp, harness side and ground as follows:

Inoperative Lamp	Connector-Pin	Circuit
LH upper	C4112 Pin 3	CLS18 (GY/BN)
LH lower	C4113 Pin 3	CLS18 (GY/BN)
RH upper	C4114 Pin 3	CLS19 (VT/OG)
RH lower	C4115 Pin 3	CLS19 (VT/OG)

- 

N0064675

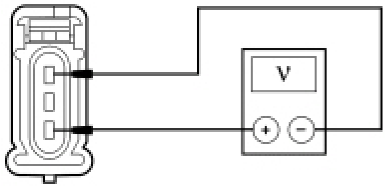
Is the voltage greater than 10 volts?

Yes	GO to J8.
No	VERIFY the BCM fuse 14 (15A) (LH stoplamps) or fuse 13 (15A) (RH stoplamps) is OK. If OK, GO to J9. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. If no DTCs are present, TEST the system for normal operation. If DTC U1000:00 is present, CLEAR the DTCs **and** REPEAT the self-test **(required to enable the stoplamp output driver if DTC U1000:00 is present).** TEST the system for normal operation. If DTC U3000:49 is present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

J8 CHECK THE REAR STOPLAMP GROUND CIRCUIT FOR AN OPEN

- While applying the brake pedal, measure the voltage between the inoperative lamp, harness side and ground as follows:

Inoperative Lamp	Connector-Pin Circuit	Connector-Pin Circuit
LH upper	C4112 Pin 3 CLS18 (GY/BN)	C4112 Pin 1 GD117 (BK)
LH lower	C4113 Pin 3 CLS18 (GY/BN)	C4113 Pin 1 GD117 (BK)
RH upper	C4114 Pin 3 CLS19 (VT/OG)	C4114 Pin 1 GD117 (BK)
RH lower	C4115 Pin 3 CLS19 (VT/OG)	C4115 Pin 1 GD117 (BK)



- N0090078

Is the voltage greater than 10 volts?

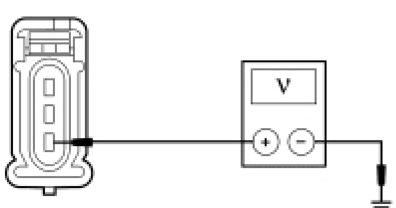
Yes	INSTALL a new bulb socket. TEST the system for normal operation.
No	REPAIR circuit GD176 (BK/WH) for an open. TEST the system for normal operation.

J9 CARRY OUT THE BCM ON-DEMAND SELF-TEST AND RECHECK FOR VOLTAGE TO THE REAR STOPLAMP

NOTE: *If both lamps are inoperative on the same side, disconnect one lamp at a time when carrying out this test.*

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: BCM Self-Test .
- Clear the DTCs and repeat the BCM on-demand self-test.
- While applying the brake pedal, measure the voltage between the inoperative lamp, harness side and ground as follows:

Inoperative Lamp	Connector-Pin	Circuit
LH upper	C4112 Pin 3	CLS18 (GY/BN)
LH lower	C4113 Pin 3	CLS18 (GY/BN)
RH upper	C4114 Pin 3	CLS19 (VT/OG)
RH lower	C4115 Pin 3	CLS19 (VT/OG)



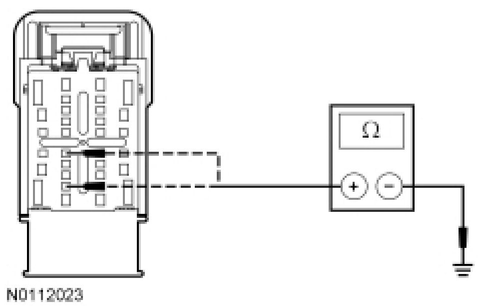
- N0064675

Is the voltage greater than 10 volts?

Yes	INSTALL a new bulb socket. TEST the system for normal operation.
No	GO to J10.

J10 CHECK THE REAR STOPLAMP VOLTAGE SUPPLY CIRCUIT FOR A SHORT TO GROUND

NOTE: *It is important to disconnect the corresponding trailer tow stop/turn relay for correct test results.*

- Ignition OFF.
- Disconnect: BCM C2280D .
- Disconnect: Corresponding Trailer Tow Stop/Turn Relay .
- Measure the resistance between the BCM C2280D Pin 12 (LH stoplamps), circuit CLS18 (GY/BN), harness side and ground; or between the BCM C2280D Pin 15 (RH stoplamps), circuit CLS19 (VT/OG), harness side and ground.
- 

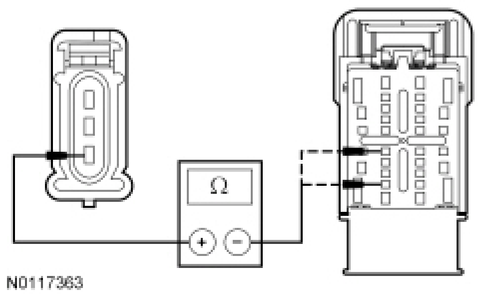
Is the resistance greater than 10,000 ohms?

Yes	GO to J11.
No	REPAIR circuit CLS18 (GY/BN) (LH stoplamps) or circuit CLS19 (VT/OG) (RH stoplamps) for a short to ground. After the repair: If no DTCs are present, TEST the system for normal operation. If DTC U1000:00 is present, CLEAR the DTCs **and** REPEAT the self-test **(required to enable the stoplamp output driver if DTC U1000:00 is present).** TEST the system for normal operation. If DTC U3000:49 is present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.

J11 CHECK THE REAR STOPLAMP VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Measure the resistance between the inoperative rear lamp, harness side and the BCM , harness side as follows:

Inoperative Lamp Connector-Pin	BCM Connector-Pin	Circuit
LH upper C4112 Pin 3	C2280D Pin 12	CLS18 (GY/BN)
LH lower C4113 Pin 3	C2280D Pin 12	CLS18 (GY/BN)
RH upper C4114 Pin 3	C2280D Pin 15	CLS19 (VT/OG)
RH lower C4115 Pin 3	C2280D Pin 15	CLS19 (VT/OG)



Is the resistance less than 5 ohms?

Yes	GO to J12.
No	REPAIR circuit CLS18 (GY/BN) (LH stoplamps) or circuit CLS19 (VT/OG) (RH stoplamps) for an open. TEST the system for normal operation.

J12 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test K: The Stoplamp(s) Are On Continuously

Refer to Wiring Diagrams Cell 90 , Turn Signal/Stop/Hazard Lamps for schematic and connector information.

Normal Operation

When the brake pedal is applied, the stoplamp switch routes voltage to the PCM and to the Body Control Module (BCM) to indicate that the brake pedal is applied.

The BCM provides voltage to all the stoplamps (on independent circuits) when it detects that the brake pedal is applied.

The voltage from the stoplamp switch also passes through the BCM fuse 31 (5A) and is routed to the Trailer Brake Control (TBC) module on one circuit and for a customer access use on another circuit.

DTC Description	Fault Trigger Conditions
• B132B:15 — Left Stop/Turn Lamp: Circuit Short To Battery or Open	This DTC sets when the BCM detects a short to voltage from the LH rear stop/turn lamp voltage supply circuit.
• B132C:15 — Right Stop/Turn Lamp: Circuit Short To Battery or Open	This DTC sets when the BCM detects a short to voltage from the RH rear stop/turn lamp voltage supply circuit.
• C0040:12 — Brake Pedal Switch "A": Circuit Short To Battery	This DTC sets when the BCM detects a short to voltage from the stoplamp switch input circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Stoplamp switch
- TBC module
- PCM
- BCM

PINPOINT TEST K : THE STOPLAMP(S) ARE ON CONTINUOUSLY

K1 DETERMINE IF ALL THE STOPLAMPS ARE ON

- Ignition ON.
- Observe the stoplamps.

Are all the stoplamps on?

Yes	GO to K3.
No	GO to K2.

K2 CHECK THE STOPLAMP VOLTAGE SUPPLY CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM C2280D .
- Ignition ON.

Does any stoplamp continue to illuminate?

Yes	If the LH rear lamps continue to illuminate, REPAIR circuit CLS18 (GY/BN) for a short to voltage. TEST the system for normal operation. If the RH rear lamps continue to illuminate, REPAIR circuit CLS19 (VT/OG) for a short to voltage. TEST the system for normal operation.

	If the high mounted stoplamp continues to illuminate, REPAIR circuit CLS52 (BN) for a short to voltage. TEST the system for normal operation.
No	GO to K9.

K3 CHECK THE STOPLAMP SWITCH

- Disconnect: Stoplamp Switch C278.

Do the stoplamps continue to illuminate?

Yes	GO to K4.
No	INSTALL a new stoplamp switch. REFER to Stoplamp Switch in this section. TEST the system for normal operation.

K4 CHECK THE PCM

- Ignition OFF.
- Disconnect: PCM C1551 (3.5L), C175B (3.7L) or C1381B (5.0L/6.2L) .
- Ignition ON.

Do the stoplamps continue to illuminate?

Yes	If the vehicle is equipped with an aftermarket TBC module, GO to K5. If the vehicle is not equipped with an aftermarket TBC module, GO to K6.
No	GO to K8.

K5 CHECK THE TBC MODULE

- Ignition OFF.
- Disconnect: TBC Module C2142.
- Ignition ON.

Do the stoplamps continue to illuminate?

Yes	GO to K6.
No	INSTALL a new TBC module. TEST the system for normal operation.

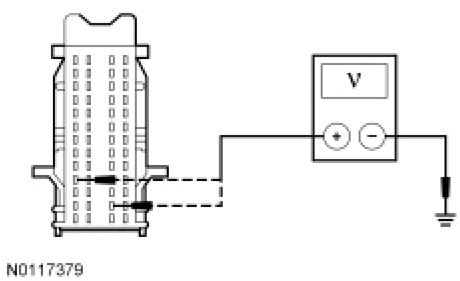
K6 CHECK THE CUSTOMER ACCESS CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM C2280F .
- Ignition ON.

Do the stoplamps continue to illuminate?

Yes	GO to K7.
No	REPAIR circuit CBP31 (BU/OG) for a short to voltage. TEST the system for normal operation.

K7 CHECK THE STOPLAMP SWITCH INPUT AND TBC MODULE BRAKE INPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM C2280B .
- Ignition ON.
- Measure the voltage between the BCM C2280B Pin 28, circuit CCB08 (VT/WH), harness side and ground; and between the BCM C2280B Pin 4, circuit CBP31 (BU/OG), harness side and ground
- 

Is any voltage present?

Yes	REPAIR circuit CCB08 (VT/WH) or circuit CBP31 (BU/OG) for a short to voltage as necessary. TEST the system for normal operation.
No	GO to K9.

K8 CHECK FOR CORRECT PCM OPERATION

- Disconnect all the PCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the PCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PCM. REFER to Section 303-14. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

K9 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.